

Chapter 1 Local Theory

These will be given in complex analysis in multivariables.

1.1 Analytic Variety

We will have a quick review of the notions/concepts in (classical) algebraic geometry:

- (Analytic Variety) A subset V of open set $U \subseteq \mathbb{C}^n$ is an analytic variety if each $p \in V$, there exists a neighborhood W such that

$$V \cap W = V(f_1, \dots, f_k)$$

Especially, an **analytic hypersurface** if V is locally a single locus of nonzero holomorphic f .

- (Irreducible Variety) We say that analytic variety V is irreducible if there exists no analytic variety V_1, V_2 such that $V_1, V_2 \neq V$, and $V_1 \cup V_2 = V$.

Now, with the notion of holomorphic functions $\mathcal{O}^n$ on

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- Discussion of Irreducibility, Decomposition in irreducible components: For $V \subseteq U \subseteq \mathbb{C}^n$, and given a hypersurface

$$V = \{z \in U \mid f(z) = 0\}$$

In a neighborhood of $0 \in V$. $\mathcal{O}_n$ is UFD, thus

$$f = \prod f_i \Rightarrow V = \cup V_i, V_i = \{z \in U \mid f_i(z) = 0\}; i = 1, \dots, k$$

Thus

♠ **Proposition 1.1.0.1** ...

◦ **Proof** ...

□

- Euclidean Algorithm:
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1.2 Complex Manifold

1.2.1 Hermitian Structure

Hermitian Structure

Definition 1.2.1.1 A **Hermitian structure** h is a sesquilinear form written

$$h = g - i\omega$$

where g is a Riemannian metric, and ω is a 2-form called a **Kähler form**, which is of type $(1, 1)$ for the almost complex structure J . As follows from the invariance of h under J :

$$J : h(u, v) = h(Ju, Jv)$$

Kähler Manifold

► Theorem ...

◦ Proof ...

□

1.2.2 Complex Manifold

Definition 1.2.2.1 ...

Liouville's Theorem

► Theorem (Liouville) Let X be a compact and connected complex manifold, then

$$\Gamma(X, \mathcal{O}_X) = \mathbb{C}$$

i.e. Any global holomorphic function on X is constant.

◦ Proof ...

□

★ **Corollary 1.2.2.2** Let X be a complex manifold of dimension at least two and $x \in X$, then

$$\Gamma(X, \mathcal{O}_X) = \Gamma(X - \{x\}, \mathcal{O}_X)$$

If X is addition compact and connected, then $\Gamma(X - \{x\}, \mathcal{O}_X) = \mathbb{C}$.

◦ Proof ...

□

Holomorphic Functions

Meromorphic Functions

The Rational Function Field

♠ Proposition 1.2.2.3 (Siegel)

◦ Proof ...

□

Some of Typical Structures

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Complex Lie Groups

1.2.3 Poincare Duality

1.3 The Projective Spaces

1.4 Blow-Ups

1.4.1 Blow-Ups on Opens

Remark Let X be a complex manifold, and $Y \subseteq X$ be a complex submanifold of codimension k . There exist holomorphic functions

$$f_1, \dots, f_k$$

with (linear) independent differentials, such that (Remark: If they have independent differentials, then $\partial f_1 \wedge \dots \wedge \partial f_k \neq 0$)

$$Y = \{z | f_i(z) = 0; i = 1, \dots, k\}$$

These equations are not unique, but we have the following:

◇ **Lemma 1.4.1.1** If $g_1, \dots, g_k$ form another system equations for Y , then locally in the neighbourhood of Y , there exists a matrix M_{ij} of holomorphic functions such that

$$g_i = \sum_j M_{ji} f_j$$

◦ **Proof ...**

□

Definition 1.4.1.2 (Blow Up of Open Sets) Let U be an open set of X , on which there exists functions

$$f_1, \dots, f_k$$

with independent differentials such that $Y \cap U = \{z \in U | f_i(z) = 0; i = 1, \dots, k\}$. Then, if we set

$$\tilde{U}_Y = \{(Z, z) \in \mathbb{P}^{k-1} \times U | Z_i f_j(z) = Z_j f_i(z), \forall i, j \leq k\}$$

where $Z = (Z_i) \in \mathbb{P}^{k-1}$,

◇ **Lemma 1.4.1.3**

- Such $\hat{U}_Y$ is a smooth complex submanifold of $\mathbb{P}^{k-1} \times U$.
- We have a map

$$\tau = pr_2 : \hat{U}_Y \rightarrow U$$

which is an isomorphism on $U - Y \cap U$.

- Its inverse being given by

$$z = (z_1, \dots, z_n) \mapsto ((f_1(z), \dots, f_k(z)), z)$$

- Above $Y \cap U$, the fibre of τ is equal to $\mathbb{P}^{k-1}$

◦ **Proof ...**

□

◇ **Lemma 1.4.1.4** Let U, V be two open sets of X , in which Y defined by equations:

$$f_1^U, \dots, f_k^U; f_1^V, \dots, f_k^V$$

respectively, with independent differentials. Then, if

$$\tau_U : \tilde{U}_Y \rightarrow U \quad \tau_V : \tilde{V}_Y \rightarrow V$$

are blowups of U and V along $U \cap V$ and $V \cap Y$ respectively, there exists a natural isomorphism

$$\phi_{UV} : \tau_U^{-1}(U \cap V) \cong \tau_V^{-1}(U \cap V)$$

◦ **Proof ...**

□

1.4.2 Blow-Ups on Manifolds

Definition 1.4.2.1 ...

Kähler Manifold

1.4.3 Line Bundles

1.5 Fubini-Study Metric

Chapter 2 Kahler Manifolds

2.1 Definitions

2.1.1 Kähler Conditions

2.2 Kähler Metrics

2.2.1 Definition and Basic Properties

2.2.2 Hermitian and Kähler Metrics

2.2.3 Basic Properties

2.3 Characterization of Kähler Metrics

2.4 Kähler Metrics and Connections

2.4.1 Metrics

Complexification

(More Details in LA.)

Let $V \in Vect_{\mathbb{R}}$, $W := Hom(V, \mathbb{R})$, then

$$W_{\mathbb{C}} := Hom_{\mathbb{R}}(V, \mathbb{C})$$

There exists decomposition

$$W_{\mathbb{C}} = W^{1,0} \oplus W^{0,1}$$

And $W_{\mathbb{R}}^{1,1} := W^{1,1} \cap \wedge^2 W_{\mathbb{R}}$, we then have

◇ **Lemma 2.4.1.1** ...

◦ **Proof** ...

□

Definition 2.4.1.2 ...

Hermitian and Kähler Metrics

2.4.2 Characterisation of Kähler Metrics

Connections

Definition 2.4.2.1 Recall: A **connection** on a real differentiable vector bundle E of rank k and class C^{∞} , which makes it possible to differentiate the sections of E . i.e. an $\mathbb{R}$ -linear map

$$\nabla : C^{\infty}(E) \rightarrow A^1(E)$$

where $A^1(E)$ is the space of the C^∞ sections of the bundle

$$\Omega_{X,\mathbb{R}} \otimes E = \text{Hom}(T_X, E)$$

♠ **Proposition 2.4.2.2** If (M, g) is a Riemannian manifold, there exist a **unique connection**

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on the tangent bundle T_M satisfying the properties:

- ∇ is compatible with g ;
- ∇ is without torsion, satisfies

...

◦ **Proof ...**

□

Chapter 3 Vector Bundles

3.1 Holomorphic Vector Bundle

3.2 Divisor

3.2.1 Analytic Hypersurface Case

Take the following notion: For hypersurface $H \subseteq U$, the family of functions

$$\{f \in \mathcal{O}_n(V) | V \in \text{Open}(\mathbb{C}^n), V \cap H = V(f)\}$$

Such function f is called **local definition function** for V near p . And this is unique up to by multiplying a nonzero functions at p .

Definition 3.2.1.1 If V has the irreducible decomposition by analytic hypersurfaces

$$V = \cup_{i=1}^k V_i$$

A **divisor** D on M is a locally finite formal linear combination

$$D = \sum a_i \cdot V_i$$

Local finite: For any $p \in M$, there exists a neighborhood of p meeting only a finite number of the V_i 's appearing in D .

$$\#\{V_i | V_i \text{ appears in } D, p \in V_i\} < \infty$$

Effective Divisor

Definition 3.2.1.2 Especially, a divisor $D = \sum a_i V_i$ is called **effective** if

$$a_i \geq 0$$

Take the following notion

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Order

Definition 3.2.1.3 ...

♠ **Proposition 3.2.1.4** There exists a natural isomorphism

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Divisor of Meromorphic Function

3.2.2 Line Bundle

Bertini's Theorem

► **Theorem** (Bertini, Complex Geometry Version) The generic element of a linear system is smooth away from the base locus of the system.

◦ **Proof ...**

□

Chern Classes of Line Bundles

3.3 Vanishing Theorems, and Corollaries

The Kodaira Vanishing Theorem

3.4 Algebraic Varieties

3.4.1 Analytic and Algebraic Varieties

3.4.2 Degree of Varieties

3.4.3 Tangent Space to Algebraic Varieties

3.5 The Kodaira Embedding Theorem

3.6 Grassmannians

Chapter 4 The Topology of Algebraic Varieties

Chapter 5 Algebraic Cycles

Chapter 6 Riemann Surfaces, and Algebraic Curves

Given in 'Riemann Surface'

Chapter 7 Surfaces

Chapter 8 Residues

Chapter 9 The Quadric Line Complex

Chapter 10 The Hodge Decomposition

Chapter 11 Variations of Hodge Structure

Chapter 12 Cohomology

12.1 Dolbeault-Cohomology